

Informatics 141

Computer Science 121

Information Retrieval

Lecture 6

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Information Retrieval

Search Engine Optimization

Coercion Techniques

Keyword Stuffing

- 1st- gen search engines relied **heavily** on textual content and frequency of words

Keyword Stuffing

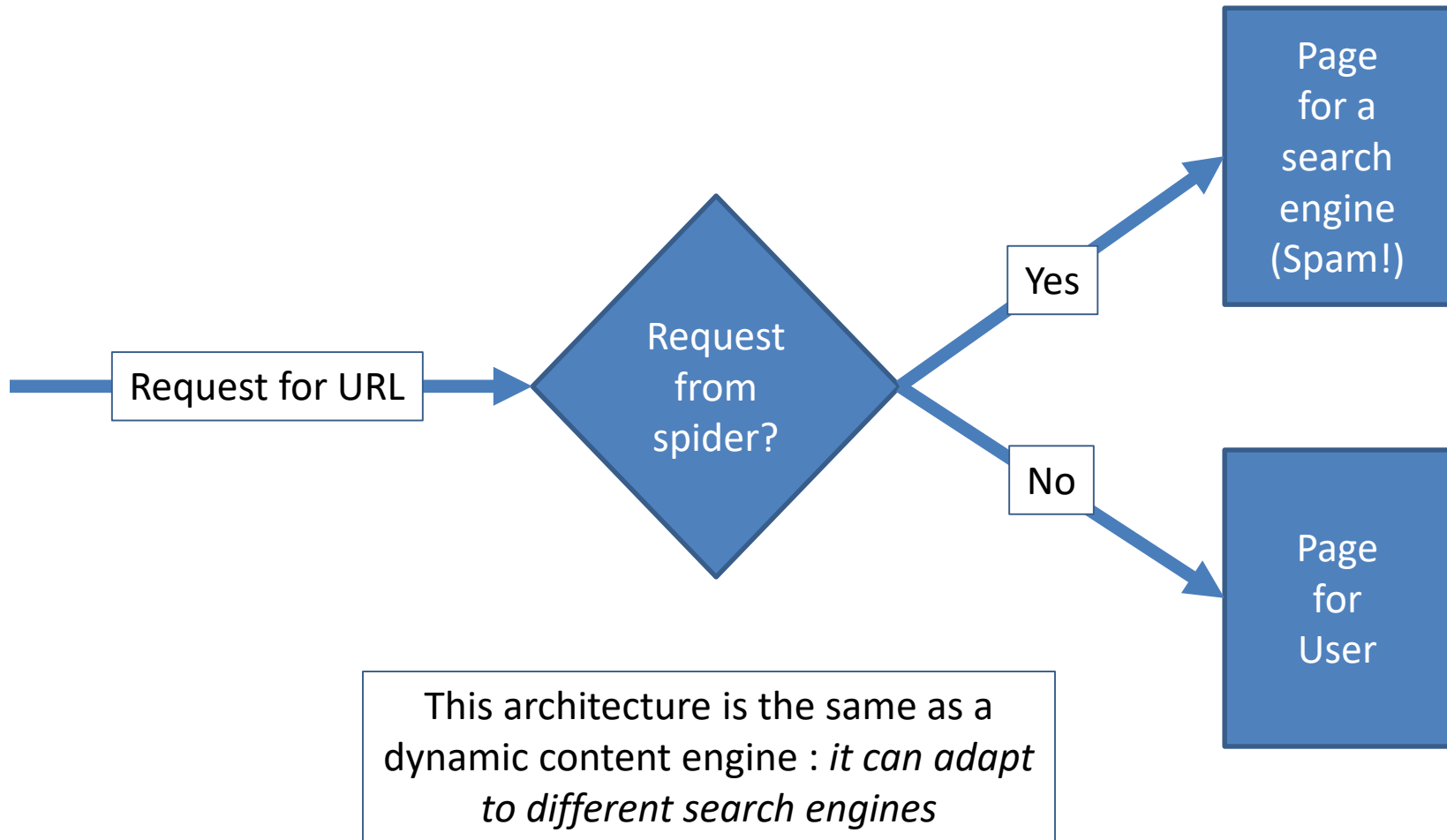
- 1st- gen search engines relied heavily on textual content and frequency of words
- SEO moved to play around with keywords
 - Misleading meta-tags

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.d
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<COMMENT TITLE="MONITOR"></COMMENT>
<meta http-equiv="Content-Language" content="en-us" />
<meta http-equiv="Content-type" content="text/html; charset=iso-8859-1"/>
<META NAME="ROBOTS" CONTENT="NOODP"><meta name="verify-v1" content="aeVxP6zTHeQzT620ipj5+ikXd/VXcdlKoYUJ/C6vVdY=" />
<META NAME="keywords" content="Expedia, Travel, Cheap Airfare, Car, Hotels, Vacations, Airfare, Car Rental, Cruises,
<META NAME="description" content="Purchase airline tickets, make hotel reservations, find vacation packages, car rent
```

- Repeating words over and over
- Playing games with colors (white on white)
 - visible to spiders/crawlers and indexers, but not users

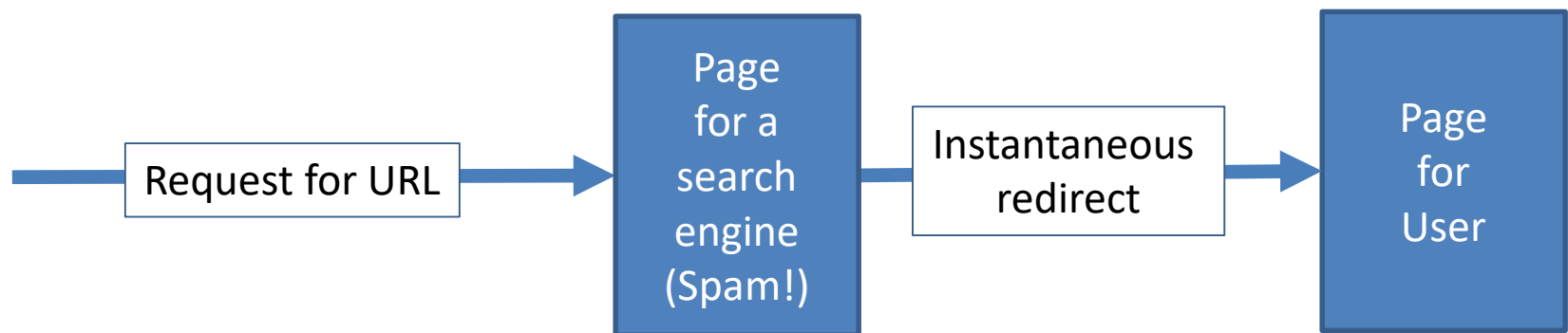
Cloaking

- Serving different content to a spider than to a browser



Doorway Pages

- Like cloaking but using a redirect (302)
 - Code 302 : “The requested resource resides **temporarily** under a different URI. Since the redirection might be altered on occasion, the client SHOULD continue to use the Request-URI for future requests”
 - **Initial page is optimized for spider**, then redirect takes user to actual content; **the user sees the final page.**



Link exchanges

- I link to you, you link to me
- “Translations”
- Universities and professors are targets
 - Essentially, due to “trust” transfer (important at the link analysis level)

Information Retrieval

Search Engine Optimization

SPAM

Two major types

- Link spamming
 - Bots that search for blogs and leave comments with links
- Clicker bots
 - Bots that issue queries and “click” on targeted query results

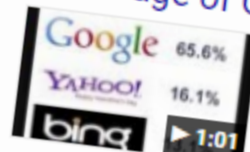
Spam industry

The First Page of Google and How I Get my Clients There

www.malleeblue.com/1st-page-google-optimization-tips/ ▼
★★★★★ Rating: 10/10 - 275 reviews

Nov 8, 2016 - Can I get your website or page to rank on the first page of Google? ... If you want to get 1st page of Google search engine results listing, make ...

First Page of Google GUARANTEED !! | How To Get Your Business At ...



https://www.youtube.com/watch?v=qjASS_NslL8 ▼
Dec 26, 2013 - Uploaded by Spotlight Ventures

Business on The First Page of Google GUARANTEED !! or Goto: ... video, blog, or Business Listing on the ...

How To Get On The First Page of Google In 24 hours - YouTube



https://www.youtube.com/watch?v=20_pdMbjR3k

May 15, 2016 - Uploaded by Amazing Tricks World

how to get first page rank on google how to get first rank in google search how to rank first on google how ...

Spamming Web contests

- Do you want to win a web contest?
 - <https://www.wholewhale.com/tips/how-to-win-or-cheat-any-online-voting-contest/>

The war on spam

- Quality indicators
 - Statistical analysis of links
 - Statistical analysis of votes

The war on spam

- Quality indicators
 - Statistical analysis of links
 - Statistical analysis of votes
- Usage indicators
 - Analytics

The war on spam

- Quality indicators
 - Statistical analysis of links
 - Statistical analysis of votes
- Usage indicators
 - Analytics
- Anti-bot mechanisms
 - Captchas



The war on spam

- Limits on meta keywords
- Family-friendly filters
- Robust link analysis
 - Ignore statistically improbable links:
 - A page in Portuguese from Portugal links to a page in Chinese from China
 - Detect cycles
 - Use link analysis to detect spammers
 - Very simple rule : Guilt by association
- Editorial (human!) intervention
 - Black-listing and creation of training sets for supervised ML
- Spam recognition by machine learning



- Search engines have SEO policies
 - What is allowed and not allowed
- **Must not be ignored**
 - Once a site is **blacklisted by a search engine**, it will virtually **disappear** from the Web

SEO “tricks” are volatile

- Dependent on how the web search engines work
- The methods behind the engines change
 - Google: PageRank (1996 - ?), Panda (2011), Penguin (2012), Hummingbird (2013), Pigeon (2014), ...
 - **Words and links.**

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 - **Words and links.**
 - Since 2016, it was using RankBrain: machine learning based.
 - **Concepts.**

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 - Google: PageRank (1996 - ?), Panda (2011), Penguin (2012), Hummingbird (2013), Pigeon (2014), ...
 - **Words and links.**
 - Since 2016, it was using RankBrain: machine learning based.
 - **Concepts.**
 - Since October 2019, Google is using BERT: NLP, machine learning.
 - Now Google Search uses : RankBrain + BERT + ... > 200 methods&signals (!)
 - Can you optimize for BERT? Google says no (but too early to tell).

SEO “tricks” are volatile

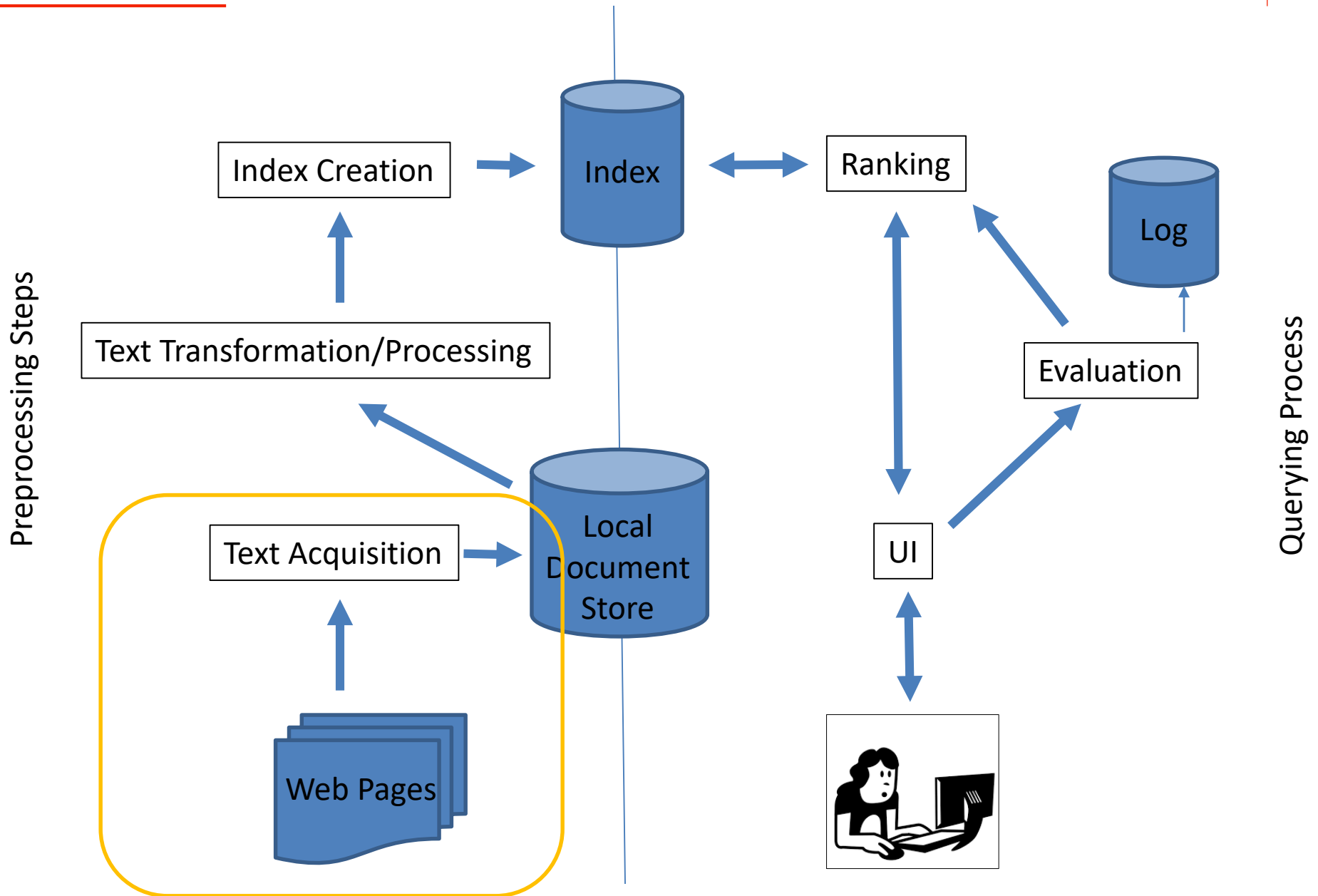
- Solution :

Optimizing for users will guarantee to optimize for good search engines

Acquiring data from the Web

Information Retrieval

Architecture



Universal Resource Identifiers

- Universal Resource Identifier (URI)
 - DEF: A string of characters used to identify a resource
- Examples of URIs:
 - <http://www.ics.uci.edu> (URL)
 - ISBN 0-486-27777-3 (URN)
 - <ftp://ftp.ics.uci.edu> (URL)

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 - <ftp://ftp.ics.uci.edu> (URL)
- URL (locator) vs URN (name)
 - Locator: must specify *where* the resource is. Name: just *what* it is.
- We are going to focus on URLs
 - But “URI” might slip in as synonym

Anatomy of a URL

- Syntax:
 - scheme://domain:port/path?query_string#fragment_id
 - authority*
 - *(the entire picture is slightly more complicated than this)*
- Full spec:
 - <http://www.w3.org/Addressing/URL/url-spec.txt>

Anatomy of a URL

- Syntax: **mandatory** **optional**

- `scheme://domain:port/path?query_string#fragment_id`
authority

- (slightly more complicated than this)

- Full spec:

- <http://www.w3.org/Addressing/URL/url-spec.txt>

Anatomy of a URL

- `http://www.ics.uci.edu/~lopes`

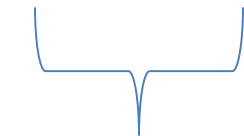
*on a web
server*

*no port!
just domain*

path

- Domains and subdomains:

– `calendar.ics.uci.edu`



Domain name

Anatomy of a URL

- `http://www.ics.uci.edu/~lopes`

*on a web
server*

*no port!
just domain*

path

query

- `http://calendar.ics.uci.edu/calendar.php?type=month&calendar=1&category=&month=02&year=2013`

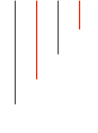
- Domains and subdomains:

– `calendar.ics.uci.edu`



Domain name

Different Flavors of Web Data Collection



- How to acquire data?
 - Data dumps
 - URL downloads
 - Web APIs
 - Web Crawling

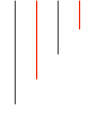
Data dumps

- Sites may package their data periodically and provide it as a “dump”
 - Example: [Wikipedia](#) (it suggests you to Torrent)
 - arXiv Bulk Full-Text Access: https://arxiv.org/help/bulk_data_s3

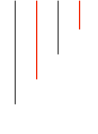
URL Downloads

- Two step process:
 1. Find out the URLs of specific resources
 2. Run a downloader that takes that list and downloads the resources
- Example: “crawling” (!) sourceforge / github for source code
- Doesn't need to be source code; can be papers, pages, etc.
 - http://link.springer.com/chapter/10.1007/978-3-642-34213-4_1
 - http://link.springer.com/chapter/10.1007/978-3-642-34213-4_2
 - ...
- Some sites use regular URLs. E.g. Google Code
 - <http://code.google.com/p/python-for-android/downloads/list>
 - ...

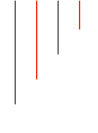
- Sites may provide (REST) interfaces for getting their data
 - Usually higher-level: avoids having to parse HTML
 - Usually restrictive: only part of the data
- Examples:
 - [Facebook Graph API](#)
 - [My data in facebook api](#)
 - [More examples](#)
 - [Youtube API](#)
 - [Twitter API](#)
 - [arXiv API](#)
 - ...



- Like people, getting HTML pages and other documents and discovering new URLs as it goes
 - Good for **changing** collections
 - Good for **unknown** documents



- Like people, getting HTML pages and other documents and discovering new URLs as it goes
 - Good for changing collections
 - Good for unknown documents
- Web admins don't like crawlers
 - Crawlers consume resources that are meant for people
 - More on this later...



Next class!

Algorithm?

- Knuth's definition of algorithm
 - **Finiteness** : *does it ends after a finite number of steps?*
 - **Definiteness** : *are the steps rigorously and unambiguously specified?*
 - **Input** : *does it has zero or more inputs?*
 - **Output** : *does it has one or more outputs related to the inputs?*
 - **Effectiveness** : *are the operations basic enough for someone to perform them using pencil and paper only?*